Business Problem

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

1. Identify underperforming brands that require promotional or pricing adjustments.

2. Determine top vendors contributing to sales and gross profit.

3. Analyze the impact of bulk purchasing on unit costs.

4. Assess inventory turnover to reduce holding costs and improve efficiency.

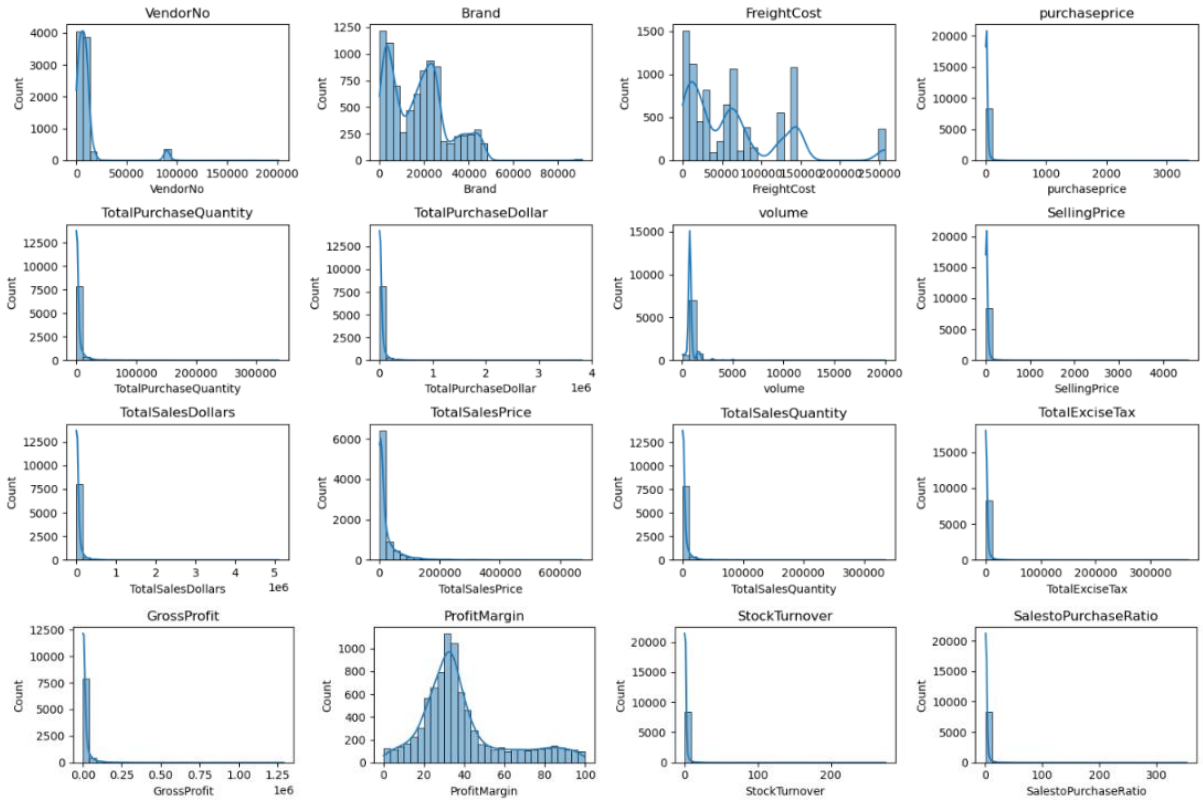
5. Investigate the profitability between high-performing and low-performing

vendors.

Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNo	10692.000000	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	201359.000000
Brand	10692.000000	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	90631.000000
FreightCost	10692.000000	61433.763216	60938.458015	0.090000	14069.870026	50293.619843	79528.990326	257032.069523
purchaseprice	10692.000000	24.385303	109.269376	0.360000	6.840000	10.455000	19.482500	5681.810059
TotalPurchaseQuantity	10692.000000	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	337660.000000
TotalPurchaseDollar	10692.000000	30106.693354	123067.799573	0.710000	453.457510	3655.464971	20738.244730	3811251.598572
volume	10692.000000	847.360550	664.309212	50.000000	750.000000	750.000000	750.000000	20000.000000
SellingPrice	10692.000000	35.643671	148.246018	0.490000	10.990000	15.990000	28.990000	7499.990234
TotalSalesDollars	10692.000000	42239.074317	167655.266047	0.000000	729.219982	5298.044970	28396.914894	5101919.546913
TotalSalesPrice	10692.000000	18793.783517	44952.773491	0.000000	289.709993	2857.799950	16059.562242	672819.341167
TotalSalesQuantity	10692.000000	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	334939.000000
TotalExciseTax	10692.000000	1774.226267	10975.582217	0.000000	4.800000	46.570001	418.650006	368242.797294
GrossProfit	10692.000000	12132.380963	46224.338053	-52002.778536	52.920004	1399.640050	8660.199938	1290667.948341
ProfitMargin	10514.000000	-15.885227	447.289882	-23730.639302	15.353839	30.778377	40.210967	99.716657
StockTurnover	10692.000000	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	274.500000
SalestoPurchaseRatio	10692.000000	2.504390	8.459068	0.000000	1.153729	1.436894	1.665449	352.928581



Negative & Zero Values:

Gross Profit: Minimum of -52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Has a minimum of – ∞, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

Freight Cost: Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

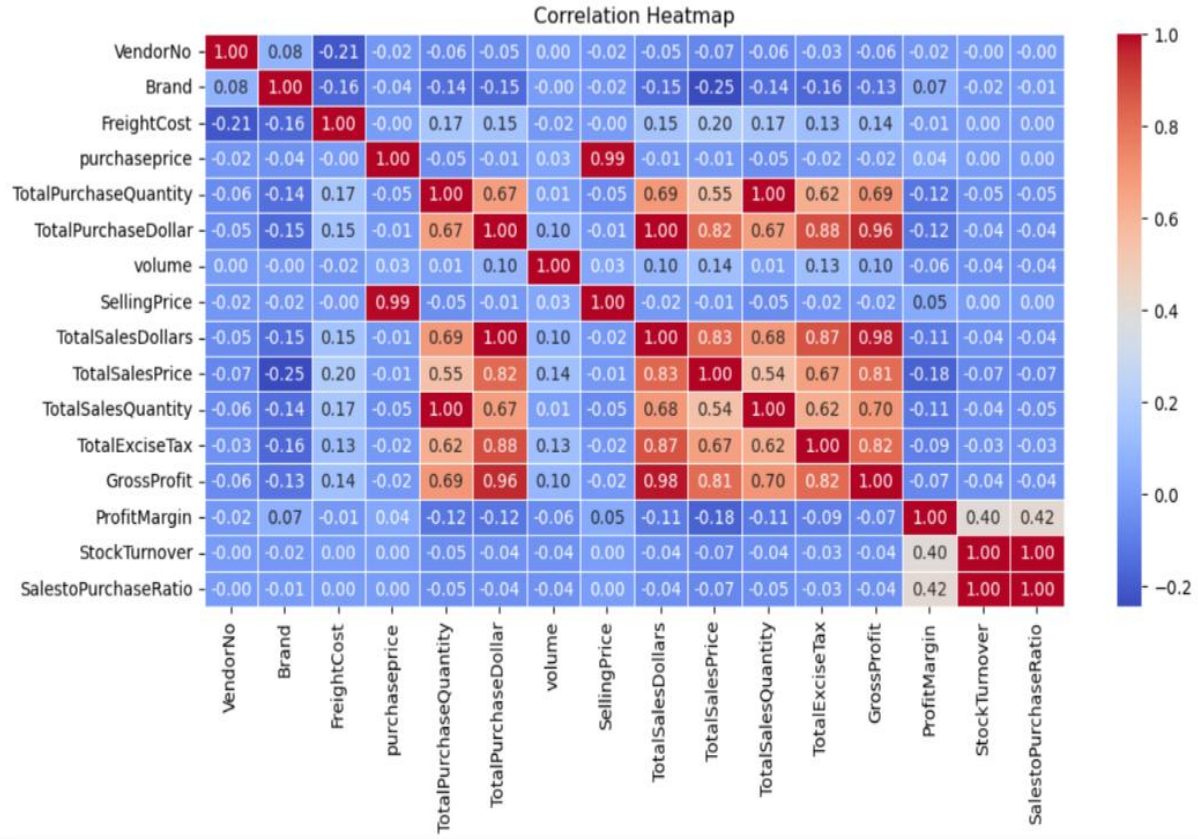
Stock Turnover: Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses).
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

Correlation Insights



Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover

Profit Margin vs. Total Sales Price: Negative correlation (-0.179) suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Research Questions & Key Findings

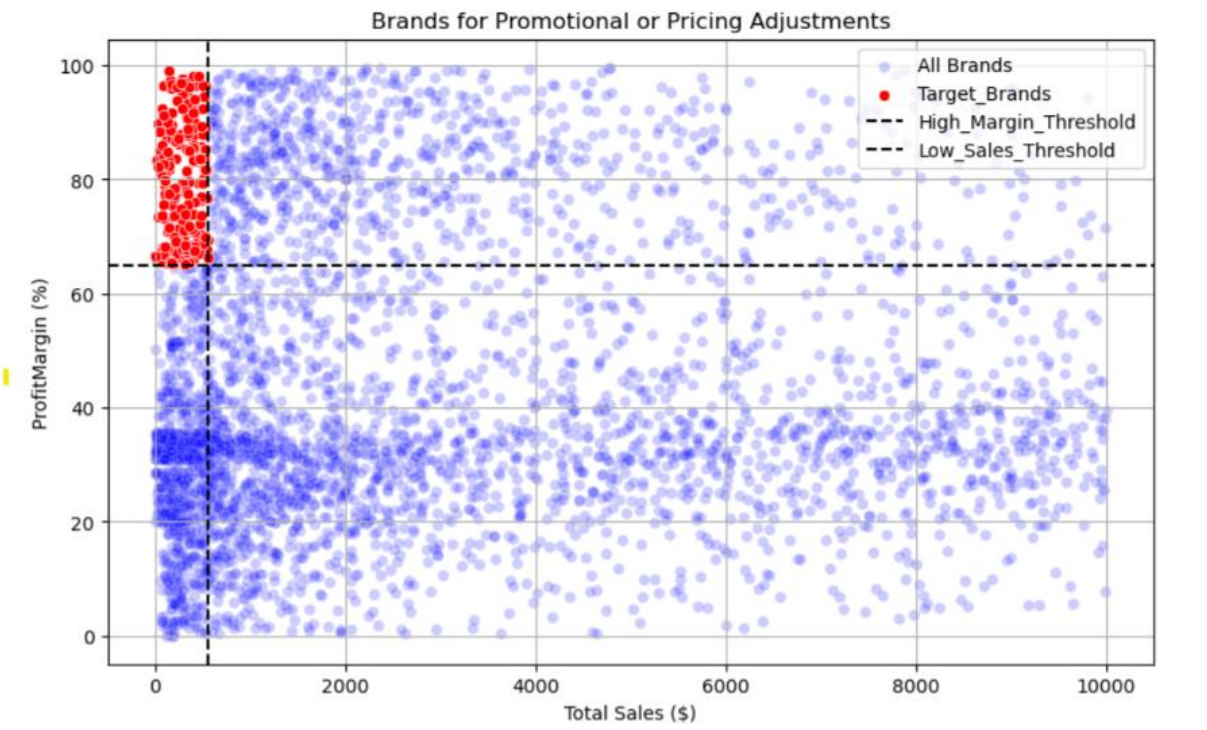
1.Brands for Promotional or Pricing Adjustments

Brands with Low Sales but High Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.990000	66.466467
2369	Debauchery Pnt Nr	11.580000	65.975820
2070	Concannon Glen Ellen Wh Zin	15.950000	83.448275
2188	Crown Royal Apple	27.859999	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.959999	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.679998	76.747312
2271	Dad's Hat Rye Whiskey	538.890007	81.851583
57	A Bichot Clos Marechaudes	539.939995	67.740861
6245	Sbragia Home Ranch Merlot	549.750000	66.444748
3326	Goulee Cos d'Estournel 10	558.870010	69.434752

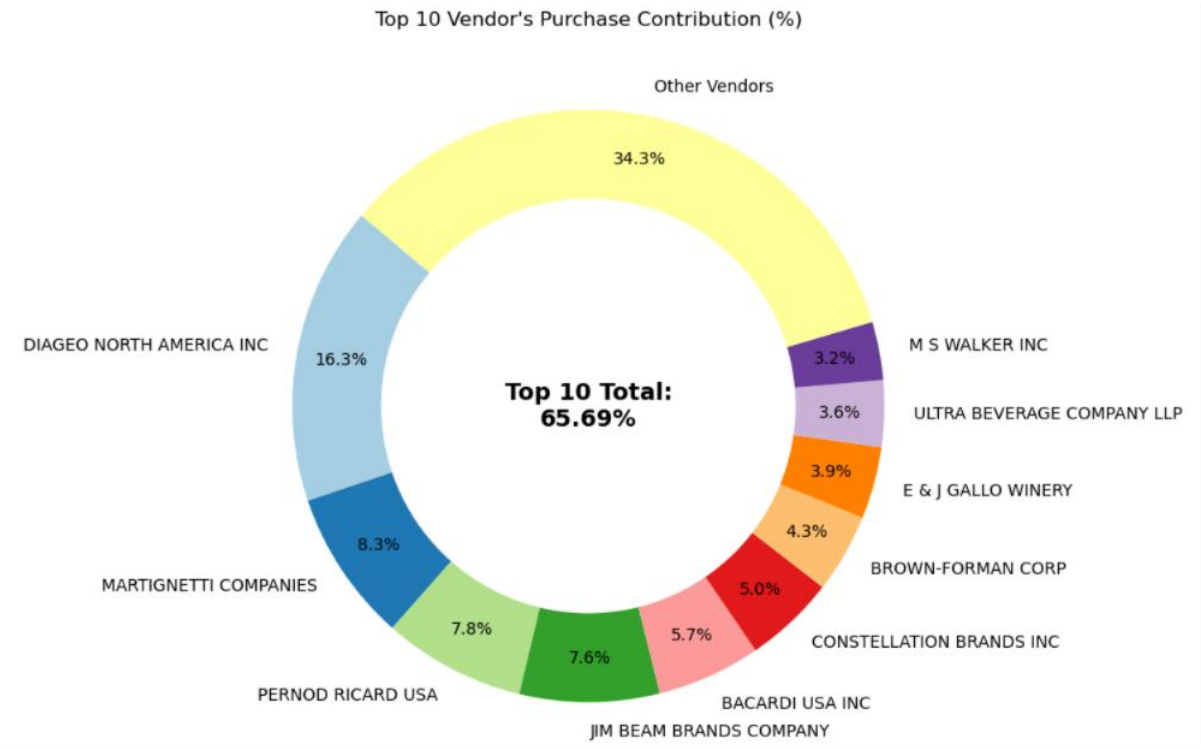
198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.



2. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost ($10.78 per unit vs. higher unit costs in smaller orders),.

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability

OrderSize	UnitPurchasePrice
Small	39.0575
Medium	15.4864
Large	10.7776

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: $2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

	vendorname	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21k
46	JIM BEAM BRANDS COMPANY	554.67k
68	PERNOD RICARD USA	470.63k
116	WILLIAM GRANT & SONS INC	401.96k
30	E & J GALLO WINERY	228.28k
79	SAZERAC CO INC	198.44k
11	BROWN-FORMAN CORP	177.73k
20	CONSTELLATION BRANDS INC	133.62k
61	MOET HENNESSY USA INC	126.48k
77	REMY COINTREAU USA INC	118.60k

5. Profit Margin Comparison: High vs. Low-Performing Vendors

Top Vendors' Profit Margin : 31.17% (Mean)

Low Vendors' Profit Margin : 41.55% (Mean)

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

```
Avg_top_vendors_profit_margin = top_vendors.mean()
print("Average Profit Margin of Top Vendors:",Avg_top_vendors_profit_margin)
```

```
Average Profit Margin of Top Vendors: 31.172242371573912
```

```
Avg_low_vendors_profit_margin = low_vendors.mean()
print("Average Profit Margin of low Vendors:",Avg_low_vendors_profit_margin)
```

```
Average Profit Margin of low Vendors: 41.54976311611542
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Actionable Insights:

• Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.

• Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.

Final Recommendations

• Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.

• Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.

• Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.

• Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.

• Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.

• By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.